SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Hackathon

Course Code:	**DSA0404**	**Course Title:**	**FUNDAMENTALS OF DATA SCIENCE**
CO Assessed:	**CO5 – AT3**	**Assessment:**	**ASSESSMENT TOOL 3 – Hackathon**
Weightage:	**40% of CIA**	**Total Marks:**	**100**
CO5 Statement:	CO5 : To understand the concept of supervised and unsupervised methods in machine learning		
Student Name:	Sanjay Kumar S	**Reg. No.:**	192324150
Section / Batch:	2023	**Date:**	31/08/2026

Code of Conduct

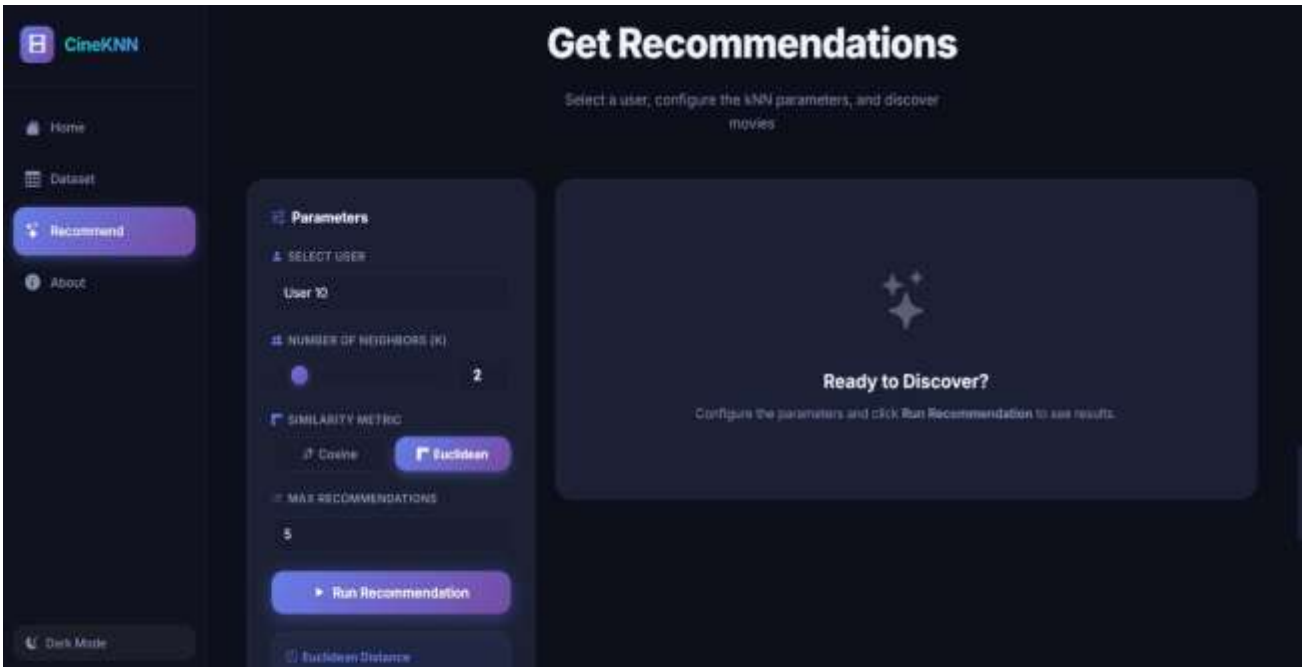
I Sanjay Kumar S / 192324150 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

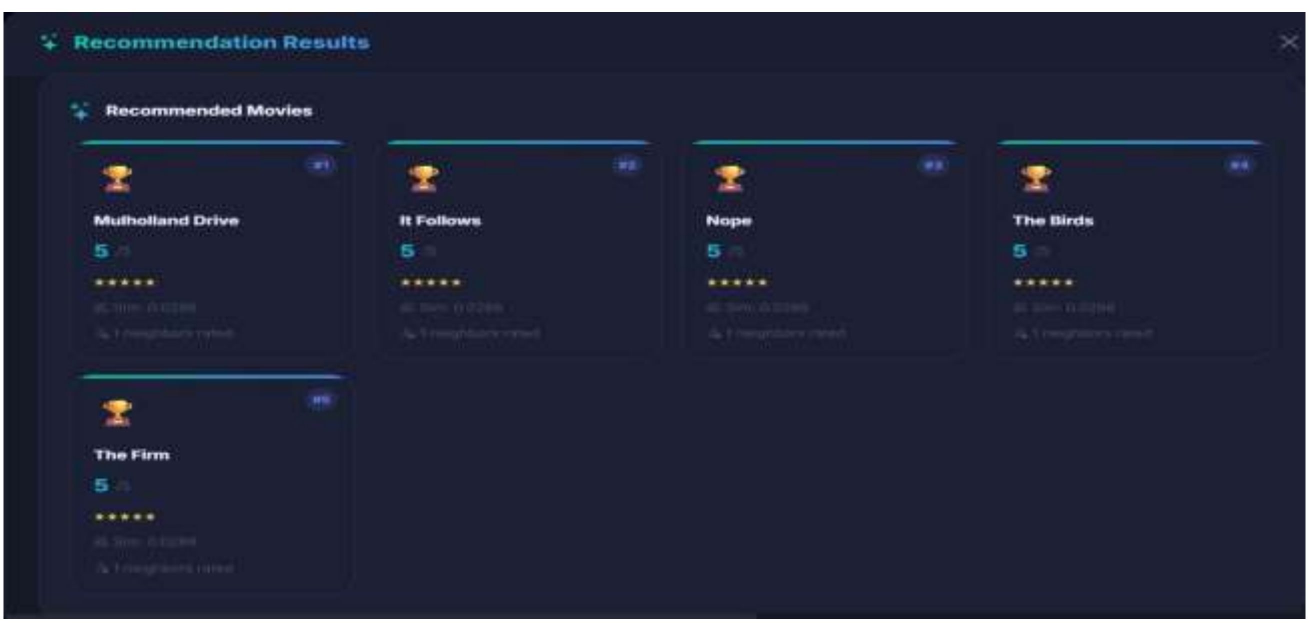
Signature of the Student:

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, wellstructured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Problem Statement

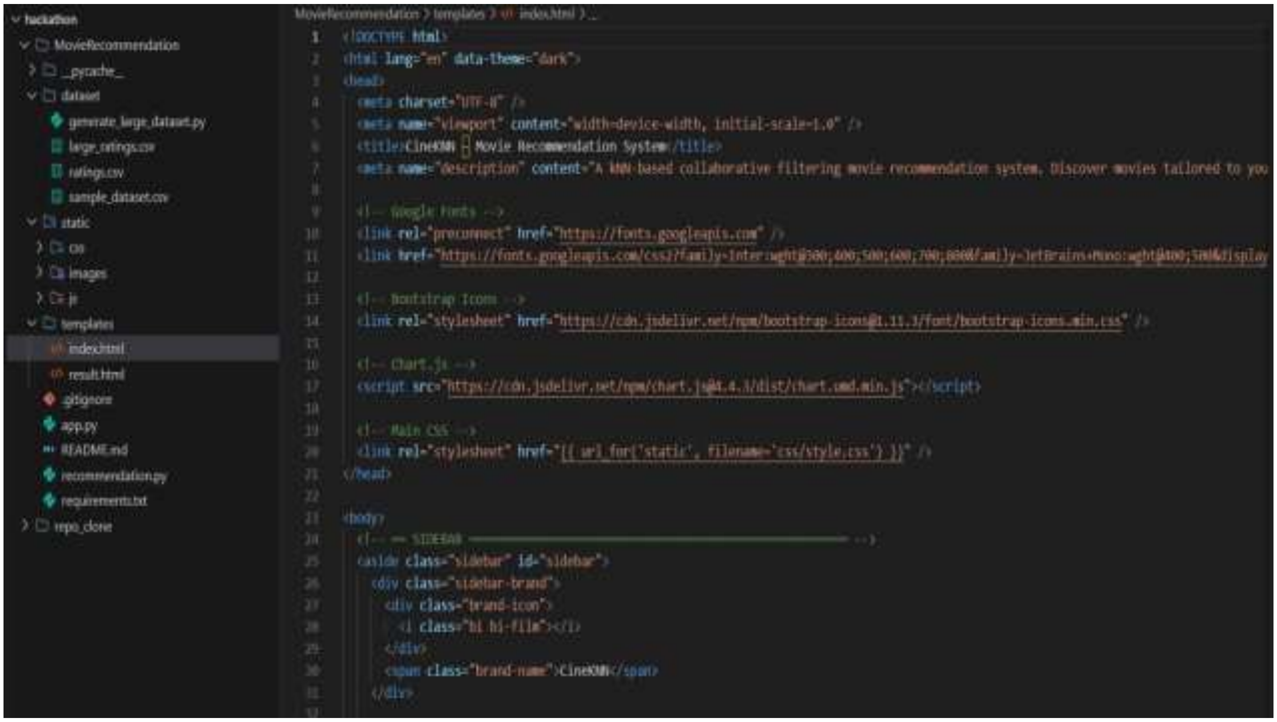
Movie/Product Recommendation via Similarity. Use kNN to recommend movies or products to a user based on similarity in ratings/features to other users (basic collaborative filtering).

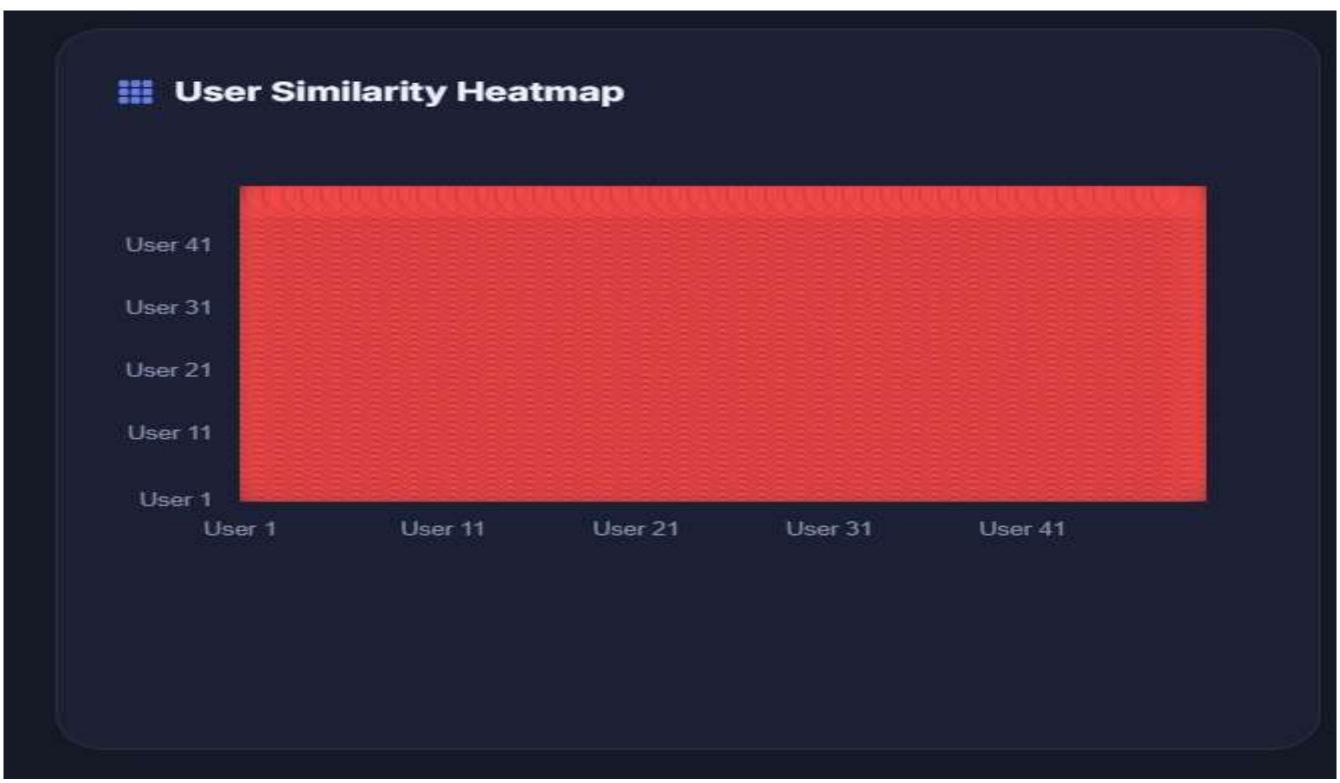





Recommendation Results
RECOMMENDATION SUMMARY
User 10 · k=2 · Euclidean Distance
2 Neighbors Found
5 Movies Recommended
PRECISION@K
100.0%
Relevant among recommended
RECALL@K
1.1%
Relevant items surfaced
MSE
3.4
Prediction error
Similar Users (k=2 Nearest Neighbors)
U8
User 8
Similarity: 0.0288
U45
User 45
Similarity: 0.0285


Recommendation Results
Predicted Rating – Bar Chart
5
4
3
2
1
0
Mulholland Drive
It Follows
Nope
The Birds
The Firm


hackathon
MovieRecommendation
pycache
dataset
generate_large_dataset.py
large_ratings.csv
ratings.csv
sample_dataset.csv
static
css
images
js
templates
index.html
result.html
.gitignore
app.py
README.md
recommendation.py
requirements.txt
repo_clone
MovieRecommendation > templates > index.html > ...
<!DOCTYPE html>
<html lang="en" data-theme="dark">
<head>
<meta charset="UTF-8" />
<meta name="viewport" content="width=device-width, initial-scale=1.0" />
<title>CineKNN – Movie Recommendation System</title>
<meta name="description" content="A KNN-based collaborative filtering movie recommendation system. Discover movies tailored to you
<!-- Google Fonts -->
<link rel="preconnect" href="https://fonts.googleapis.com" />
<!-- Bootstrap Icons -->
<link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.11.3/font/bootstrap-icons.min.css" />
<!-- Chart.js -->
<script src="https://cdn.jsdelivr.net/npm/chart.js@4.4.3/dist/chart.umd.min.js"></script>
<!-- Main CSS -->
<link rel="stylesheet" href="{{ url_for('static', filename='css/style.css') }}" />
</head>
<body>
<!-- == SIDEBAR ==== -->
<aside class="sidebar" id="sidebar">
<div class="sidebar-brand">
<div class="brand-icon">
<i class="bi bi-film"></i>
</div>
CineKNN
</div>


User Similarity Heatmap
User 41
User 31
User 21
User 11
User 1
User 1
User 11
User 21
User 31
User 41


GPS Map Camera
Google
Saveetha Nagar
NH 48, Kuthambakkam, Chembarambakkam, Tamil Nadu
Lat 13.024309° Long 80.023239°
Monday, 31/08/2026 11:04 GMT +05:30